VACF Running-Diffusion Plotting Specification

VP2: Green-Kubo D(t) Visualization

mdstats

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1 Purpose

This document specifies

`mdstats/plotting/vacf_transport.py`

and its public function

`plot_vacf_diffusion(result, ...)`

The function visualizes one or more already computed `VACFDiffusionResult` objects. It does not recompute a VACF, perform numerical quadrature, fit a long-time tail, choose a plateau, or declare a converged transport coefficient.

VP2 is implemented in `mdstats 0.19.9a0`.

2 Physical background

For an isotropic system in d spatial dimensions, the running Green-Kubo self-diffusion integral is

$$D(t) = \frac{1}{d} \int_0^t \langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(\tau) \rangle \, d\tau.$$

For one Cartesian direction α,

$$D_\alpha(t) = \int_0^t \langle v_{i\alpha}(0) v_{i\alpha}(\tau) \rangle \, d\tau.$$

The time-correlation transport framework follows Green [1] and Kubo [2]. The input result and cumulative trapezoidal quadrature are owned by `integrate_vacf_to_diffusion()` and its analysis specification. VP2 only renders the stored finite-time running integral.

The limit

$$D = \lim_{t \to \infty} D(t)$$

is meaningful only when a stable long-time regime exists and the trajectory is sufficiently sampled. Oscillation, return toward zero, or negative finite-time values can be physically meaningful for trapped or vibrational motion. The plotter therefore never labels the final sample as an asymptotic diffusion coefficient.

3 Provenance

3.1 Borrowed software machinery

Rendering uses Matplotlib, cited to Hunter [3].

3.2 Reused mdstats scientific machinery

The plotter consumes only `VACFDiffusionResult`, whose numerical and physical contract is defined by the Green-Kubo transport module. Unit conversion from $\text{Å}^2/\text{ps}$ to cm^2/s is already exposed by that result type.

3.3 mdstats-specific design

The following are mdstats decisions:

- one plotting function for a single result, sequence, or label mapping;
- explicit time and diffusion units;
- no automatic plateau selection or tail fitting;
- an optional zero reference line;
- deterministic default labels for scalar and Cartesian curves;
- returning `(Figure, Axes)` without calling `show()` or saving files.

4 Public API

```
from collections.abc import Mapping, Sequence
from typing import Literal

from matplotlib.axes import Axes
from matplotlib.figure import Figure
```

```python
def plot_vacf_diffusion(
    result: (
        VACFDiffusionResult
        | Sequence[VACFDiffusionResult]
        | Mapping[str, VACFDiffusionResult]
    ),
    *,
    ax: Axes | None = None,
    time_unit: Literal["fs", "ps", "ns"] = "ps",
    diffusion_unit: Literal["angstrom2/ps", "cm2/s"] = "cm2/s",
    label: str | None = None,
    labels: Sequence[str] | None = None,
    title: str | None = None,
    show_zero_line: bool = True,
    grid: bool = True,
) -> tuple[Figure, Axes]:
    ...
```

5 Input contract

Every plotted object must be a validated `VACFDiffusionResult` containing:

```
lag_times
running_diffusion_a2_per_ps
running_diffusion_cm2_per_s
component
dimensions
weighting
integration
```

The plotter does not accept arbitrary arrays because the result type carries the physical weighting and unit guarantees needed for a correct label.

6 Multiple-result contract

Accepted forms are:

```
plot_vacf_diffusion(na_result)
plot_vacf_diffusion([na_result, k_result], labels=["Na", "K"])
plot_vacf_diffusion({"Na": na_result, "K": k_result})
```

For one result, `label=` may be supplied. For a sequence, `labels=` must have the same length. For a mapping, its keys are authoritative labels and `labels=` is invalid.

7 Unit contract

7.1 Time

The stored time axis is in ps. Display conversion is

$$t_{\mathrm{fs}} = 10^{3} t_{\mathrm{ps}}, \qquad t_{\mathrm{ns}} = 10^{-3} t_{\mathrm{ps}}.$$

7.2 Diffusion

The canonical stored running integral is in $\mathrm{Å}^2/\mathrm{ps}$. The displayed SI-derived unit uses

$$1\ \text{Å}^2/\text{ps} = 10^{-4}\ \text{cm}^2/\text{s}.$$

Unit changes affect plotted copies only and never mutate the result.

8 Rendering contract

The function:

1. resolves and validates labels;
2. converts the time coordinate for display;
3. selects the requested diffusion unit;
4. draws one line per result;
5. optionally draws a light $D = 0$ reference;
6. sets labels, title, grid, and legend when needed;
7. returns the owning `Figure` and `Axes`.

It does not smooth, interpolate, truncate, normalize, or resample the curve.

9 Example workflow

The package example

`examples/vacf_dynamics/vasp_contcar_trajectory_vdos_diffusion.py`

reads a watcher-generated VASP `TRAJECTORY`, computes a mass-weighted VACF-derived VDOS, computes a uniformly weighted self VACF for a selected mobile species, integrates it into $D(t)$, and saves separate VDOS and running self-diffusion figures.

For the supplied Na-LTA file:

```
python examples/vacf_dynamics/vasp_contcar_trajectory_vdos_diffusion.py \
    TRAJECTORY \
    --timestep-fs 1.0 \
    --diffusion-species Na \
    --output-dir vacf_outputs
```

10 Error handling

The function rejects:

- unsupported result types;
- empty result collections;
- non-string or empty labels;
- inconsistent sequence-label lengths;
- unsupported time or diffusion units;
- non-boolean display switches;
- invalid axes objects.

11 Tests

Focused tests verify:

1. top-level and plotting-package exports;
2. exact use of stored $\text{Å}^2/\text{ps}$ values;
3. exact conversion to cm^2/s;
4. fs, ps, and ns horizontal axes;
5. mapping and sequence labels;

6. optional zero reference;
7. no mutation of result arrays;
8. invalid-option failures;
9. real-file generation of separate VDOS and $D(t)$ figures.

12 References

[1] M. S. Green, "Markoff random processes and the statistical mechanics of time-dependent phenomena. II. Irreversible processes in fluids," *Journal of Chemical Physics* **22**, 398-413 (1954). DOI: 10.1063/1.1740082.

[2] R. Kubo, "Statistical-mechanical theory of irreversible processes. I. General theory and simple applications to magnetic and conduction problems," *Journal of the Physical Society of Japan* **12**, 570-586 (1957). DOI: 10.1143/JPSJ.12.570.

[3] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering* **9**, 90-95 (2007). DOI: 10.1109/MCSE.2007.55.